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20 January 1969
SUPERSEDING
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24 November 1953

MILITARY SPECIFICATION

CONTROL PANEL C-826/AIC-10

1. SCOPE

1.1 This specification covers one type of control panel, designated Control Panel C-826/AIC-10. Control Panel C-826/AIC-10 is one item of Intercommunication Set AN/AIC-10.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250/8C	Aluminum Alloy, 5052, Plate and Sheet

Military

MIL-R-11	Resistor, Fixed, Composition (Insulated), General Specification for
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-E-5556	Enamel, Camouflage, Quick Drying
MIL-E-5557	Enamel, Heat-Resisting, Glyceryl-Phlhalate, Black
MIL-R-6130	Rubber, Cellular, Chemically Blown
MIL-P-6808	Primer Coating, Zinc Chromate, For Aircraft and Missile Applications, Application of
MIL-P-7788	Plate, Plastic, Lighting
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment

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STANDARDS

Federal

FED-STD-141	Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-831	Test Reports, Preparation of
MS25010	Light, Panel-Plastic Plate Lighting
MS25100	Switch, Toggle, Two Pole, Sealed Toggle
MS25212	Control Panel, Console Type, Aircraft Equipment, Base Dimensions
MS25213	Control Panel, Aircraft Equipment, Typical Installation

DRAWINGS

Air Force

52B13673	Plate, Identification, Toggle Switch
53C12843	Stand, Test, Control Panel
53C12844	Wiring Diagram, Control Panel C-826/AIC-10 Test Circuit, Schematic, Crosstalk
53C12845	Wiring Diagram, Control Panel C-826/AIC-10, Test Circuit, Schematic, Circuit Continuity
53C13646	Outline Dimensions, Control Panel C-826/AIC-10
68C35762	Control Panel - C-826/AIC-10
68C53515	Wiring Diagram, Control Panel C-826/AIC-10
68D53516	Control Panel Assembly C-826/AIC-10

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General Specification. The requirements of MIL-E-5400 for Class 2 equipment apply to equipment provided under this specification. Where the two specifications conflict the requirements of this specification shall govern.

3.2 Construction. The control panel shall be built in accordance with Drawing 68D53516.

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3.2.1 Mounting Plate. The mounting plate shall be of half hard aluminum sheet of 0.064 inch thickness, in accordance with QQ-A-250/8C, to which are mounted the plastic lighting plate, the support for the electrical receptacle, the captive fastener studs which serve to attach the control panel to the Aircraft structure, and the toggle switches and other components required. Dimensions shall be as specified by MS25212. Component fasteners, except those extending all the way through the plastic plate such as control shafts and switches, shall not extend beyond the front surface of the mounting plate.

3.2.2 Plastic Lighting Plate. The plastic lighting plate shall be made from a product qualified to MIL-P-7888. Further requirements and dimensions are shown on Drawing 68C35762.

3.2.3 Drainage. Each control panel shall incorporate adequate drainage provisions for any moisture condensing within the unit. When the control panel is mounted in any usable position (including horizontal, oblique, vertical, overhead and sidewise mounting) all closed corners, channels, etc., which would collect and hold water shall be eliminated or be provided with adequate means for drainage.

3.2.4 Wiring. The control panel shall be wired in accordance with Drawing 68C53515.

3.2.5 Cabling. The cable harness shall be fabricated so that crosstalk is at least 50db down when tested as specified in 4.6.3.

3.3 Wire. The hook-up wire shall be AWG #22 in accordance with MIL-W-16878, type B. The outer covering shall be synthetic textile of glass braid in accordance with MIL-W-16878. Wiring shall be color coded in accordance with Drawing 68C35762.

3.4 Resistors. The resistor shall by type RC20GF273K per MIL-R-11.

3.5 Toggle switch identification plate. The toggle switch identification plate shall be in accordance with Drawing 52B13673.

3.6 Panel light assembly. The light assembly shall be in accordance with MS25010.

3.7 Fasteners. Fasteners for mounting the control panel to the aircraft structure shall be in accordance with MIL-F-25173.

3.8 Toggle switches. The toggle switches shall be type MS25100-23.

3.9 Connector. The connector shall be type AN3102A-18-1P in accordance with MIL-C-5015.

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3.10 Circuit label. The circuit label shall be a reduced copy of the schematic diagram shown on Drawing 68C35762 except that the word identification of the terminals of Jack J601 shall not be included.

3.11 Dimensions. Overall dimensions shall not exceed those shown on Drawing No. 53C13646.

3.12 Weight. The weight of the control panel shall not exceed one pound.

3.13 Finish. The mounting surfaces of the control panel shall be kept free of any insulating material in order to provide a good electrical grounding surface. All aluminum parts shall be anodized in accordance with MIL-A-8625. The front face and edges of the mounting plate shall be finished with black non-reflecting enamel conforming to MIL-E-5556 or MIL-E-5557.

3.14 Identification of product. The control panel, subassemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130. The nameplate shall be located as shown on drawing no. 53C13646.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The government reserves the right to perform any of the tests or inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the control panel shall be classified as follows:

a. Preproduction tests

b. Acceptance tests

4.3 Test conditions. Unless otherwise specified the control panel shall be tested under the following conditions:

a. Ambient room temperature - plus 20°C to plus 35°C

b. Relative humidity - room ambient (20 to 90 percent relative)

c. Atmospheric pressure - ground level ambient.

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4.4 Preproduction testing.

4.4.1 Preproduction test samples. A minimum of 10 control panels shall be subjected by the contractor to the preproduction test specified herein. When specified in the contract, two control panels shall be submitted to the procuring agency with the preproduction test report. The samples will be used by the procuring agency for:

a. A review of the mechanical construction of the control panel.

b. Performance of any tests which may be deemed necessary after a review of the contractors preproduction test report.

c. Government reference purposes.

4.4.2 Preproduction test reports. After completion of the preproduction tests by the contractor, 3 copies of a complete test report shall be furnished to the procuring activity. The preproduction test report shall be prepared in accordance with MIL-STD-831. In addition, the test report shall include the name and grade of all materials used in the control panel.

4.4.3 Preproduction tests. Preproduction tests shall consist of all the following test as described under 4.6:

a. Group I

(1) Visual and mechanical inspection
(2) Electrical circuit test
(3) Crosstalk measurement test
(4) Mounting test

b. Group II

(1) Vibration tests
(2) Shock tests
(3) High temperature tests
(4) Low temperature tests
(5) Accelerated weathering

c. Group III

(1) Humidity tests
(2) Salt spray tests
(3) Fungus resistance tests

4.4.4 Preproduction test routing. All sample control panels shall be subjected to all tests in Group I. After completion of the tests in Group I, the control panels shall be divided into 5 sample lots consisting of two control panels. Each of the five sample lots shall be subjected to a different test from the list of test under Group II. After completion of tests in Group II, three sample lots shall be subjected to a different

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test from the list of test under Group III.

4.5 Acceptance tests. Acceptance shall consist of:

a. Individual tests
b. Sampling tests

4.5.1 Individual tests. Each control panel shall be subjected to the following tests as described under 4.6:

a. Visual and mechanical inspection
b. Electrical circuit test

4.5.2 Sampling tests. One control panel shall be selected at random from each lot of 25 or fraction thereof which have passed the individual tests. The selected control panel shall be subjected to the following tests:

a. Crosstalk measurement
b. Mounting test

4.5.2.1 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.5.2.2 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Visual and mechanical inspection. The control panel shall be examined for design construction, materials, components and workmanship to determine compliance with the requirements of this specification. Any deviations from the requirements shall be cause for rejection.

4.6.2 Electrical circuit test. Each control panel shall be tested for satisfactory operation using the test circuit and instructions shown on Drawing Nr 53012045.

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4.6.3 Crosstalk measurement test. The control panel shall be tested for satisfactory crosstalk suppression using the test circuit and instructions shown on drawing number 53C12844.

4.6.4 Mounting test. The control panel shall be tested for proper fastener operation by installing it on a suitable test stand having fastener receptacle strips rigidly mounted in accordance with MS25213.

4.6.5 Vibration tests. The control panel shall be tested in accordance with Procedure XII of MIL-E-5272 when mounted in a test stand constructed as shown by drawing number 53C12843. Variations in the length of the test stand may be made as required, but all other dimensions shall be as shown on the drawing. The vibration amplitude required by the vibration test shall be measured at the point on the test stand where the control panel is secured to the rigid upright.

4.6.6 Shock tests. The shock test shall be conducted in accordance with Procedure V of MIL-E-5272 and the control panel shall be secured to a test stand shown on drawing 53C12843.

4.6.7 High temperature tests. The high temperature test shall be conducted in accordance with Procedure II of MIL-E-5272.

4.6.8 Low temperature tests. The low temperature test shall be conducted in accordance with Procedure II of MIL-E-5272.

4.6.9 Accelerated weathering. Finished panels shall be exposed to accelerated weathering for 150 hours in accordance with FED-STD-141, Method 6152 except that water spray shall not be used. After exposure, the panel shall be visually compared to a similar unexposed panel to determine compliance with 3.2.2.

4.6.10 Humidity tests. The humidity test shall be conducted in accordance with Procedure I of MIL-E-5272.

4.6.11 Salt spray tests. The salt spray test shall be conducted in accordance with Procedure I of MIL-E-5272.

4.6.12 Fungus resistance tests. The control panel shall be tested for fungus resistance in accordance with Procedure I of MIL-E-5272.

4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be in accordance with MIL-E-17555. Levels of preservation, packaging and packing shall be specified by the procuring activity. The quantity of unit pack shall be one each control panel.

5.2 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Control Panel C-826/AIC-10 is intended for use with Control Panel C-824/AIC-10 in AN/AIC-10 intercommunication systems to increase the number of audio channels that can be monitored simultaneously by a crew member.

6.2 Ordering data. Procurement documents shall specify the following:

a. The title, number and date of this specification.

b. Preproduction test sample if required.

c. Level of preservation, packaging and packing.

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